

Supplementary Materials for MoDex: A Diffusion Policy for Sequential Multi-Object Dexterous Grasping

1 Implementation Details

1.1 RL Fine-tuning Reward

This appendix provides implementation details for the shaped reward r_t^{base} defined in the Eq. (4) of the main text and used for RL fine-tuning. Unless otherwise specified, the environment uses $r_{\text{succ}} = 9.0$, $r_{\text{min}} = -0.5$, $r_{\text{max}} = 2.0$, and $\lambda_{\text{ina}} = 0.02$. Let $\mathcal{M}_i$ denote the set of objects that must be maintained at stage i , including the current target object and all objects grasped in previous stages.

Success condition. At each environment step, we first evaluate the task-success condition that selects the success branch of the reward function. At stage i , a step is considered successful when the current target object is lifted above the table by the lift threshold ($h_{i,t} = 1$) and every object in $\mathcal{M}_i$ remains in hand-contact ($c_{j,t}^{\text{hold}} > 0$). During RL training, a rollout can last up to 500 steps and is not terminated immediately upon first satisfying this condition. Instead, the success condition is re-evaluated at every subsequent step, so the success branch $r_{\text{succ}} - \lambda_{\text{ina}} r_t^{\text{ina}}$ is paid out persistently as long as the grasp is maintained, encouraging the policy not only to lift the target object but also to keep the grasp stable throughout the remainder of the rollout.

Contact extraction. At each simulator step, contacts are extracted from MuJoCo contact pairs between the target object’s collision geoms and the Allegro hand collision geoms. The implementation separates hand contacts into fingertip contacts, non-tip finger link contacts, and palm contacts. Let $\mathcal{G}_{\text{obj}}$ denote the target object geoms and $\mathcal{G}_{\text{hand}}$ denote all hand geoms. A MuJoCo contact pair (g_a, g_b) is treated as a hand-object contact if one geom belongs to $\mathcal{G}_{\text{hand}}$ and the other belongs to $\mathcal{G}_{\text{obj}}$:

$$(g_{\text{hand}}, g_{\text{obj}}) \in \mathcal{G}_{\text{hand}} \times \mathcal{G}_{\text{obj}}. \quad (1)$$

For all such pairs, the hand-side geom is added to the contacted hand-geom set

$$\mathcal{H}_t^{\text{contact}} = \{g_{\text{hand}} \in \mathcal{G}_{\text{hand}} \mid \exists g_{\text{obj}} \in \mathcal{G}_{\text{obj}} \text{ such that } (g_{\text{hand}}, g_{\text{obj}}) \text{ is a MuJoCo contact at time } t\}. \quad (2)$$

The contacted finger set is then

$$\mathcal{F}_t^{\text{contact}} = \left\{ f \mid \mathcal{G}_f^{\text{tip}} \cap \mathcal{H}_t^{\text{contact}} \neq \emptyset \text{ or } \mathcal{G}_f^{\text{link}} \cap \mathcal{H}_t^{\text{contact}} \neq \emptyset \right\}, \quad (3)$$

where $f \in \{\text{th}, \text{ff}, \text{mf}, \text{rf}\}$. Thus, a finger is counted as contacting the target if either its fingertip or one of its non-tip finger links touches the target object. Palm contacts are counted separately as

$$n_t^{\text{palm}} = |\mathcal{G}^{\text{palm}} \cap \mathcal{H}_t^{\text{contact}}|. \quad (4)$$

The implementation also counts hand-table contacts, denoted n_t^{table} , by checking MuJoCo contacts between hand geoms and table or floor geoms.

Opposition-space contact score. The active opposition space determines a set of required contact sites $\mathcal{C}_i$. For standard finger-finger opposition spaces, $\mathcal{C}_i$ contains the active fingertips. For the ring-palm opposition space, $\mathcal{C}_i$ contains the ring fingertip and the ring-side palm support site. Each site m has a distance $d_{m,t}$ to the target object and a contact score $c_{m,t} \in [0, 1]$.

The implementation first assigns an unnormalized site weight w_m and then normalizes the weights over the required contact sites:

$$\tilde{w}_m = \frac{w_m}{\sum_{m' \in \mathcal{C}_i} w_{m'}}. \quad (5)$$

The proximity and contact scores are computed with the normalized weights:

$$p_t = \sum_{m \in \mathcal{C}_i} \tilde{w}_m \exp(-\kappa_m d_{m,t}^2), \quad c_t = \sum_{m \in \mathcal{C}_i} \tilde{w}_m c_{m,t}. \quad (6)$$

For standard finger-finger opposition spaces, each active fingertip is assigned the same unnormalized weight $w_m = 1$ and uses $\kappa_m = 80$. Since these opposition spaces contain two active fingertips, the normalized weights are

$$\tilde{w}_{f_1} = \tilde{w}_{f_2} = \frac{1}{2}. \quad (7)$$

We identify each required fingertip site with its corresponding finger label. The contact scores are binary:

$$c_{m,t} = \mathbb{I}[m \in \mathcal{F}_t^{\text{contact}}]. \quad (8)$$

Thus, for an opposition space with active fingers f_1 and f_2 ,

$$c_t = \frac{1}{2} \mathbb{I}[f_1 \in \mathcal{F}_t^{\text{contact}}] + \frac{1}{2} \mathbb{I}[f_2 \in \mathcal{F}_t^{\text{contact}}]. \quad (9)$$

For ring-palm, the unnormalized weights are $w_{\text{ring}} = 0.7$ and $w_{\text{palm}} = 0.3$, which already sum to one. Therefore, the normalized weights are unchanged:

$$\tilde{w}_{\text{ring}} = 0.7, \quad \tilde{w}_{\text{palm}} = 0.3. \quad (10)$$

The ring fingertip uses $\kappa_{\text{ring}} = 80$, while the palm support site uses $\kappa_{\text{palm}} = 28$. The ring contact score is binary:

$$c_{\text{ring},t} = \mathbb{I}[\text{rf} \in \mathcal{F}_t^{\text{contact}}]. \quad (11)$$

Using the palm-contact count n_t^{palm} defined above, the palm support score is softened as

$$c_{\text{palm},t} = \tanh(n_t^{\text{palm}}). \quad (12)$$

Therefore, the ring-palm aggregate contact score is

$$c_t = 0.7 c_{\text{ring},t} + 0.3 c_{\text{palm},t} = 0.7 \mathbb{I}[\text{rf} \in \mathcal{F}_t^{\text{contact}}] + 0.3 \tanh(n_t^{\text{palm}}). \quad (13)$$

The target grasp reward is

$$r_t^{\text{grasp}} = 0.25 p_t + 0.85 c_t. \quad (14)$$

Maintenance reward. For each object $j \in \mathcal{M}_i$, the implementation computes an object maintenance score

$$q_{j,t} = h_{j,t} c_{j,t}^{\text{hold}} s_{j,t}. \quad (15)$$

Here $h_{j,t}$ is a normalized lift-height score. Let $z_{j,t}^{\text{bottom}}$ denote the bottom height of object j and z^{table} denote the table height. The lifted height above the table is clipped to the success range $[0, h_{\text{succ}}]$ and then normalized:

$$h_{j,t} = \frac{\text{clip}(z_{j,t}^{\text{bottom}} - z^{\text{table}}, 0, h_{\text{succ}})}{h_{\text{succ}}}, \quad h_{\text{succ}} = 0.1 \text{ m}. \quad (16)$$

Thus, $h_{j,t} = 0$ when the object is at the table height and $h_{j,t} = 1$ once the object reaches the lift success threshold.

The soft hold score $c_{j,t}^{\text{hold}}$ measures whether object j is currently held by the hand. The implementation counts the number of distinct hand collision geoms that are in contact with object j , denoted $n_{j,t}^{\text{hand-obj}}$, and normalizes it by two hand-object contacts:

$$c_{j,t}^{\text{hold}} = \text{clip}\left(\frac{n_{j,t}^{\text{hand-obj}}}{2}, 0, 1\right). \quad (17)$$

Therefore, zero hand-object contacts gives $c_{j,t}^{\text{hold}} = 0$, one contact gives $c_{j,t}^{\text{hold}} = 0.5$, and two or more distinct hand-object contacts give $c_{j,t}^{\text{hold}} = 1$.

The object stability score penalizes large object motion:

$$s_{j,t} = \exp(- (0.08 \|\dot{x}_{j,t}\|_2 + 0.03 \|\omega_{j,t}\|_2)), \quad (18)$$

where $\dot{x}_{j,t}$ and $\omega_{j,t}$ are the linear and angular velocities of object j . This term is close to one for a stable held object and decreases as the object moves or rotates quickly.

The maintenance score is computed as a soft minimum over all objects that must remain maintained:

$$\text{softmin}_{j \in \mathcal{M}_i}(q_{j,t}) = -\tau \log \left(\frac{1}{|\mathcal{M}_i|} \sum_{j \in \mathcal{M}_i} \exp(-q_{j,t}/\tau) \right), \quad (19)$$

with $\tau = 0.1$. This makes the reward sensitive to the least well-maintained object while providing a smooth-shaped scalar for RL fine-tuning.

Finally, the maintenance reward is gated by the squared opposition-space contact score:

$$r_t^{\text{maintain}} = c_t^2 \text{softmin}_{j \in \mathcal{M}_i}(q_{j,t}). \quad (20)$$

The gate delays the maintenance reward until the commanded opposition-space contact is established: weak or missing OS contact gives little maintenance reward, while confident OS contact allows the object-retention term to dominate.

Avoidance. The avoidance term penalizes off-opposition-space contacts and table contacts:

$$r_t^{\text{avoid}} = r_t^{\text{off}} + r_t^{\text{table}}. \quad (21)$$

Let n_t^{inactive} be the number of non-commanded fingers contacting the target object, and let $e_t^{\text{OS}} \in \{0, 1\}$ indicate whether the commanded opposition-space contact has been established. The off-OS contact penalty is applied only before the commanded opposition space is engaged:

$$r_t^{\text{off}} = (1 - e_t^{\text{OS}}) (0.04 n_t^{\text{inactive}} + 0.04 \max(0, n_t^{\text{palm}} - n_t^{\text{palm,des}})). \quad (22)$$

where n_t^{palm} is the number of palm-target contacts and $n_t^{\text{palm,des}}$ is the desired number of palm contacts under the current opposition space.

The table penalty is

$$r_t^{\text{table}} = 0.05 n_t^{\text{table}} + 0.15 \sum_{g \in \mathcal{G}_{\text{hand}}} \max(0, 0.02 - z_{g,t}), \quad (23)$$

where $z_{g,t}$ is the height of hand collision geom g above the table. The second term softly discourages hand geoms from moving too close to or below the table surface.

Inactive-finger motion penalty. Inactive fingers are fingers that are neither part of the current commanded opposition space nor part of any previously used opposition space in the grasp history. Let $\mathcal{F}_{\text{ina}}$ denote this set, and let $\mathcal{J}_f$ denote the set of joints belonging to finger f . For each inactive finger, the implementation computes the mean absolute normalized joint deviation from the initial Allegro hand pose, together with a velocity penalty:

$$r_t^{\text{ina}} = 0.7 \frac{1}{|\mathcal{F}_{\text{ina}}|} \sum_{f \in \mathcal{F}_{\text{ina}}} \frac{1}{|\mathcal{J}_f|} \left\| \frac{q_{f,t} - q_{f,0}}{q_f^{\text{max}} - q_f^{\text{min}}} \right\|_1 + 0.3 \frac{1}{|\mathcal{F}_{\text{ina}}|} \sum_{f \in \mathcal{F}_{\text{ina}}} \tanh(\|\dot{q}_{f,t}\|_2). \quad (24)$$

This penalty enters the success branch of the reward function, so the policy only collects the full r_{succ} constant when the fingers that are not required by the current or previous grasps remain near their initial pose. Concretely, when the success condition is satisfied, $r_t^{\text{base}} = r_{\text{succ}} - \lambda_{\text{ina}} r_t^{\text{ina}} = 9.0 - 0.02 r_t^{\text{ina}}$; otherwise,

$$r_t^{\text{base}} = \text{clip}(r_t^{\text{grasp}} + r_t^{\text{maintain}} - r_t^{\text{avoid}}, -0.5, 2.0). \quad (25)$$

1.2 Hyperparameters

Table 1 summarizes the hyperparameters used to train MoDex. The policy is optimized with Adam and trained as a conditional denoising diffusion model using a DDIM noise scheduler. Observations consist of a downsampled point cloud encoded by a PointNet encoder together with low-dimensional proprioceptive state, and we inject small Gaussian noise into both modalities during training to improve robustness and narrow the sim-to-real gap.

Hyperparameter	Value
Learning rate	1×10^{-4}
Batch size	256
Noise scheduler	DDIM
Train timesteps	100
Inference timesteps	10
Observation horizon	2
Action horizon	4
Prediction horizon	8
Num. of parameters	7M
Point cloud encoder	PointNet
Point cloud feature dim	128
Number of points	1024
Point cloud noise std	0.01
Low-dim noise std	0.01

Table 1: Diffusion policy hyperparameters.

1.3 Dataset Curation Details

To enhance policy robustness and data coverage, Stage 1 randomizes the initial object pose within ± 10 cm in the x and y directions (translation) and $\pm 45^\circ$ (rotation), and adds Gaussian noise with $\sigma = 0.01$ to the Franka arm joints and $\sigma = 0.05$ to the Allegro hand joints. For Stages 2 and 3, only the initial object pose is randomized, since the rest of the environment state is restored from a successful trajectory in $\mathcal{D}_{i-1}$; joints occupied by previously grasped objects are locked during motion planning to maintain control of those objects.

2 Extended Simulation Analysis

2.1 Per-Stage and Per-OS Simulation Results

Table 2 provides a detailed breakdown of simulation success rates across three grasping stages and four opposition space conditions. While the main paper reports stage-level averages, this fine-grained analysis shows how each method behaves under different OSes. T+I, I+M, M+R, and R+P denote thumb+index, index+middle, middle+ring, and ring+palm OSes, respectively.

Overall, MoDex achieves the strongest performance across most stage–OS combinations. In Stage 1, where the robot only needs to grasp a single object, MoDex obtains high success rates on the finger–finger opposition spaces, reaching 81.7% on T+I, 95.0% on I+M, and 78.3% on M+R. Its lower success rate of 45.0% on R+P indicates that ring–palm grasps are more difficult in the single-object setting, likely because they require coordinated fingertip and palm support rather than fingertip contact alone. Even so, MoDex matches or outperforms the learning-based baselines on most Stage 1 OS conditions.

The advantage of MoDex becomes clearer in later stages. In Stage 2, the policy must grasp the current target while retaining one previously grasped object. BC-RNN degrades sharply on I+M, M+R, and R+P, suggesting that behavior cloning alone does not reliably preserve previously occupied fingers during sequential grasping. MoDex-BC remains competitive on some OS conditions,

Stage 1					
Method	T+I	I+M	M+R	R+P	Total
BC-RNN	53.33±2.36	43.33±2.36	51.67±2.36	61.67±2.36	52.50±1.02
MoDex-BC	43.33±4.71	76.67±6.24	66.67±9.43	41.67±6.24	57.08±2.57
w/o Grasp History Context	41.67±10.27	83.33±4.71	78.33±4.71	46.67±4.71	62.50±3.68
MoDex (full)	81.67±2.36	95.00±4.08	78.33±8.50	45.00±10.80	75.00±3.68
DPPO w/o r_t^{grasp}	68.33±9.43	90.00±7.07	83.33±4.71	41.67±6.24	70.83±1.18
DPPO w/o r_t^{maintain}	66.67±8.50	90.00±4.08	86.67±6.24	36.67±6.24	70.00±3.68

Stage 2					
Method	T+I	I+M	M+R	R+P	Total
BC-RNN	35.00±8.16	3.33±2.36	6.67±2.36	1.67±2.36	11.67±2.12
MoDex-BC	80.00±8.16	30.00±4.08	15.00±4.08	75.00±4.08	50.00±3.06
w/o Grasp History Context	80.00±10.80	36.67±10.27	23.33±4.71	55.00±10.80	48.75±5.30
MoDex (full)	83.33±6.24	35.00±7.07	15.00±7.07	65.00±7.07	49.58±1.18
DPPO w/o r_t^{grasp}	81.67±12.47	35.00±7.07	21.67±10.27	51.67±10.27	47.50±2.04
DPPO w/o r_t^{maintain}	76.67±12.47	26.67±4.71	21.67±10.27	51.67±12.47	44.17±4.12

Stage 3					
Method	T+I	I+M	M+R	R+P	Total
BC-RNN	46.67±6.24	10.00±4.08	18.33±10.27	35.00±4.08	27.50±2.70
MoDex-BC	71.67±14.34	38.33±8.50	20.00±4.08	38.33±10.27	42.08±1.56
w/o Grasp History Context	68.33±8.50	25.00±4.08	20.00±7.07	21.67±10.27	33.75±2.70
MoDex (full)	65.00±10.80	35.00±4.08	21.67±2.36	58.33±2.36	45.00±2.70
DPPO w/o r_t^{grasp}	66.67±16.50	31.67±6.24	31.67±8.50	23.33±2.36	38.33±6.64
DPPO w/o r_t^{maintain}	58.33±16.50	33.33±6.24	26.67±8.50	51.67±8.50	42.50±1.77

Table 2: Per-stage and per-opposition space (OS) grasp success rates (%) in simulation. Results are reported as mean $\pm$ standard deviation over three seeds. The Total column reports the average success rate across all four OS conditions for the corresponding stage.

especially T+I, but its performance is less consistent across morphologies. In contrast, MoDex maintains strong performance on T+I and R+P, reaching 83.3% and 65.0%, respectively. This suggests that the combination of OS conditioning, grasp-history context, and RL fine-tuning helps the policy allocate the hand’s kinematic resources more effectively.

Stage 3 is the most constrained setting, as the hand must grasp a third object while maintaining two previously grasped objects. Under this setting, MoDex continues to outperform the baselines on most OS conditions, achieving 65.0% on T+I, 35.0% on I+M, and 58.3% on R+P. The relatively strong R+P result in Stage 3 is particularly important: although ring-palm opposition is harder in Stage 1, it becomes useful when other fingers are already occupied, because the palm can provide additional support without requiring a full two-finger opposition pattern. This supports the motivation of using OS-conditioned policies for sequential multi-object grasping, where different grasp morphologies provide different trade-offs depending on the remaining available fingers.

PPO fails in all evaluated conditions, which confirms that training a dexterous sequential grasping policy from scratch is highly challenging due to the high-dimensional action space and sparse long-horizon success structure. BC-RNN and MoDex-BC achieve reasonable performance in easier settings but degrade or vary substantially across later-stage OS conditions. Taken together, these results show that MoDex not only improves average success rate, but also provides more reliable OS-specific behavior across increasingly constrained sequential grasping stages.

2.2 Why SeqDiffuser Fails: Kinematic Infeasibility

SeqDiffuser [1] frequently produces grasp poses that satisfy contact and force-closure criteria on the object in isolation, but place the hand in configurations that the Franka–Allegro system cannot reach without violating joint limits or colliding with the table. As shown in Fig. 1, the synthesized

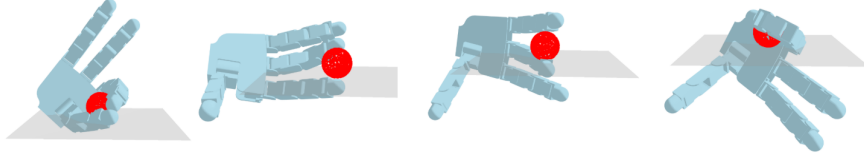


Figure 1: Representative SeqDiffuser grasp samples on Stage 1. The target object is shown in red, the Allegro hand in blue, and the table surface in grey.

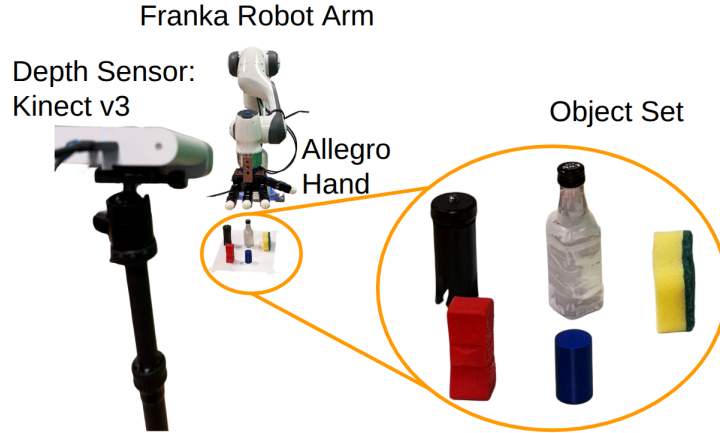


Figure 2: Experimental setup. An Allegro Hand attached to a Franka Emika Panda arm and a Kinect v3 depth sensor. The object set comprises three everyday objects (a tripod, a bottle, and a sponge) and two 3D-printed primitives (a red block and a blue cylinder).

hand pose often hovers in a pose that would require the arm to pass through the table surface to realize. Because grasp synthesis is decoupled from arm kinematics, these infeasible candidates are only filtered downstream by the motion planner, which then returns failures.

3 Real-World Experiments: Detailed Results

3.1 Real-World Hardware and Object Set

The real-world experimental setup is shown in Fig. 2. We use an Allegro Hand mounted on a Franka Emika Panda arm, with scene perception provided by a Kinect v3 depth sensor, whose extrinsics are calibrated using an ArUco marker [2]. The object set consists of five objects: three everyday items (a tripod, a bottle, and a sponge) and two 3D-printed primitives (a red block and a blue cylinder). Only the blue cylinder is seen during training. We do not evaluate any thumb-related opposition space (i.e. T+I) in the real-world experiments due to a hardware limitation of the thumb. As illustrated in Fig. 3, we ran a simple diagnostic in which a zero action command was issued throughout the motion: under gravity alone, the thumb was unable to support its own weight and collapsed, driving the arm into a collision and triggering a robot error (reflex) state. We therefore exclude thumb-based opposition spaces from the real-robot evaluation and report results only for the remaining OS conditions.

3.2 Extended Real-World Results

Table 3 reports the per-object outcomes underlying the aggregate real-world success rates in the main text, with three trials per object per stage. Following the hardware limitation discussed in Section 3.1, the thumb-based opposition space (T+I) is excluded. While the main text reports stage-

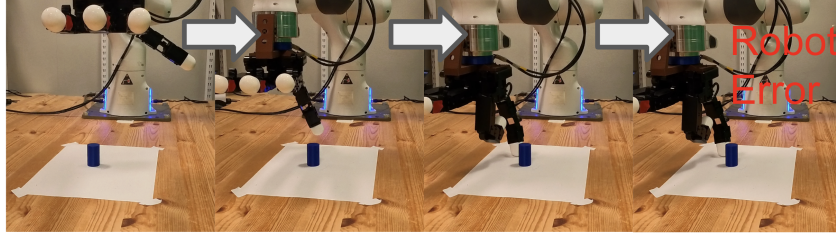


Figure 3: Hardware limitation of the thumb. Under a zero-action command, the thumb cannot withstand gravity, causing a collision and triggering a robot reflex error.

Method	Stage	OS	Cylinder	Box	Sponge	Tripod	Bottle	Success (%)
MoDex	1	I+M	2/3	2/3	2/3	2/3	2/3	66.7
		M+R	2/3	2/3	2/3	1/3	1/3	53.3
		R+P	1/3	2/3	2/3	2/3	1/3	53.3
	2	I+M	0/3	0/3	1/3	1/3	3/3	33.3
		M+R	0/3	0/3	0/3	2/3	1/3	20.0
	3	I+M	0/3	0/3	2/3	0/3	1/3	20.0
MoDex-BC	1	I+M	0/3	1/3	1/3	2/3	1/3	33.3
		M+R	1/3	2/3	2/3	3/3	1/3	60.0
		R+P	0/3	0/3	2/3	1/3	1/3	26.7
	2	I+M	0/3	0/3	1/3	0/3	2/3	20.0
		M+R	0/3	0/3	1/3	1/3	1/3	20.0
	3	I+M	0/3	0/3	1/3	0/3	0/3	6.67

Table 3: Per-object trial outcomes and success rates (%) for MoDex and MoDex-BC across stages and OS conditions. Each entry reports the number of successful trials out of three.

level averages, this fine-grained view exposes how performance is distributed across objects and opposition spaces, and how the sim-to-real gap evolves across stages.

Table 4 reveals a consistent difficulty ordering that holds for both MoDex and MoDex-BC: the everyday objects (sponge, tripod, and bottle) are grasped most reliably, whereas the rigid 3D-printed primitives (cylinder and box) are the hardest. Notably, the blue cylinder is the only object *seen* during training, yet it is among the lowest-performing objects in the real world. This indicates that, under real sensing and contact dynamics, graspability is governed more by an object’s geometry and surface properties than by whether it appeared in the training distribution. The compressible sponge tolerates imprecise finger closure and offers high surface friction, while the tripod and bottle present thin, wrappable features (legs and a neck) that afford stable finger-finger opposition. In contrast, the smooth low-friction cylinder tends to roll out of a forming grasp, and the rigid box demands precise, tight closure on flat faces with no compliance to absorb small placement errors. This object-level structure is consistent with the sim-to-real gap being driven primarily by contact-level effects rather than by perception alone.

Table 5 summarizes a matched sim-to-real comparison, where success rates are averaged over the OS conditions actually evaluated on hardware at each stage, excluding the thumb-based T+I space. Under this matched comparison, MoDex exhibits a small and roughly stable sim-to-real gap: about 15% in Stage 1 and Stage 3, and no degradation in Stage 2, where the real-world success rate slightly exceeds its simulation counterpart. MoDex-BC, by contrast, degrades more severely at the hardest stage, with its gap widening to roughly 32% in Stage 3, while MoDex retains a 15% gap. Consequently, the margin of MoDex over MoDex-BC, which is already present in Stage 1, widens on real hardware as the task becomes more constrained. This per-stage view sharpens the main-text observation: DPPO fine-tuning does not merely provide a uniform performance offset; rather,

Method	Cylinder	Box	Sponge	Tripod	Bottle	Overall
MoDex	27.8	33.3	50.0	44.4	50.0	38.9
MoDex-BC	5.6	16.7	44.4	38.9	33.3	27.8

Table 4: Per-object grasp success rates (%) in the real-world experiments.

Method	Stage / OSes	Simulation	Real world	Gap
MoDex	Stage 1 / I+M, M+R, R+P	72.78	57.78	15.00
	Stage 2 / I+M, M+R	25.00	26.67	-1.67
	Stage 3 / I+M	35.00	20.00	15.00
MoDex-BC	Stage 1 / I+M, M+R, R+P	61.67	40.00	21.67
	Stage 2 / I+M, M+R	22.50	20.00	2.50
	Stage 3 / I+M	38.33	6.67	31.66

Table 5: Matched sim-to-real comparison over the OS conditions evaluated on hardware. Success rates are averaged over the listed OS conditions for each stage. The gap is computed as simulation minus real-world success rate.

it reduces the rate at which performance collapses as the sequential-grasping task becomes more constrained, precisely the regime in which behavior cloning alone leaves the policy most brittle.

3.3 OS-Planning Prompt for Multi-Object Stages

For the multi-object stages (Stage 2 and Stage 3), we query Gemini [3] to plan the grasp sequence. Given the list of objects in the scene, the designated target object, and the target opposition space f_i to use for it, the model decides which of the remaining objects to grasp in the earlier stages and which opposition space to assign to each. The model receives a short scene description and an RGB image of the scene, and returns a structured plan that respects the hand’s kinematic constraints. At inference time, the returned plan is parsed into the per-stage (object, f_i) sequence executed by the policy. An example text prompt is:

SYSTEM

You are a planning assistant for a dexterous robot hand that grasps multiple objects one at a time without releasing the objects it is already holding. The hand is an Allegro Hand with four fingers: thumb (T), index (I), middle (M), and ring (R). A grasp is defined by an Opposition Space (OS): the pair of contacts that squeeze the object. The available opposition spaces are:

- thumb + index finger
- index + middle finger
- middle + ring finger
- ring finger + palm

RULES

1. The hand can hold at most 3 objects in total.
2. Each opposition space may be used at most once across the whole sequence (a finger committed to one object cannot be reused for another).
3. Objects already held must stay held; only currently free fingers may be assigned to the next object.
4. The target object must be grasped LAST, using exactly the given target opposition space.
5. For the earlier stages, choose which of the remaining objects to grasp and with which opposition space, so that the target opposition space is still free when the target is reached.
6. Do NOT use the thumb+index (T+I) opposition space for any object; only I+M, M+R, and R+P are allowed.

INPUT

Available objects: {object_list}.
 Target object: {target_object}.
 Target opposition space: {target_os}.
 The RGB image of the scene.

TASK

Output the full grasping plan as an ordered JSON list, where the final stage grasps the target object with the target opposition space and the earlier stages grasp the chosen preceding objects. Each element has the form:

```
{ "stage": <int>, "object": "<name>", "os": "<OS label>" }
```

Return ONLY the JSON list, with no extra text.

EXAMPLE OUTPUT

```
[
  { "stage": 1, "object": "bottle", "os": "R+P" },
  { "stage": 2, "object": "sponge", "os": "M+R" },
  { "stage": 3, "object": "cylinder", "os": "I+M" }
]
```

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